ALGEBRAIC SPACES OVER S ARE EQUIVALENT TO PAIRS OVER S' MAPPING TO S

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays two matrix blocks describing the two categories; they are part of the counted statement span and are preserved verbatim.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The change of base lemma for localization functors. Lemma lemma-change-base-scheme (2949-2979, produced as L44175), used at proof lines 3090-3093. Makes (j_!F, j_!F -> S) an object of the right hand side.
- j_! induces an equivalence of categories of sheaves. Sites, Lemma sites-lemma-essential-image-j-shriek, used at proof lines 3094-3096 and 3108-3110. Reduces the equivalence to the single implication proved.
- Presentation of an algebraic space. Lemma lemma-space-presentation (1387-1407), used at proof lines 3098-3101. Supplies U' and R'.
- The presentation theorem for algebraic spaces. Theorem theorem-presentation, used at proof lines 3110-3112. Concludes that F = U/R is an algebraic space.

Source reference. Original lemma 03I4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4590–4590. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0262: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2835–2835. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00Y1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1239–1239. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 02WW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3819–3819. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let Sch_{fppf} be a big fppf site. Let $S \to S'$ be a morphism of this site. The construction above give an equivalence of categories*

$$\left\{\begin{array}{c}\text{category of algebraic}\\ \text{spaces over } S\end{array}\right\} \leftrightarrow \left\{\begin{array}{c}\text{category of pairs } (F', F' \to S) \text{ consisting}\\ \text{of an algebraic space } F' \text{ over } S' \text{ and a}\\ \text{morphism } F' \to S \text{ of algebraic spaces over } S'\end{array}\right\}$$

Proof. Let F be an algebraic space over S. The functor from left to right assigns the pair $(j_!F, j_!F \to S)$ to F which is an object of the right hand side by Lemma 03I4. Since this defines an equivalence of categories of sheaves by Sites, Lemma 00Y1 to finish the proof it suffices to show: if F is a sheaf and $j_!F$ is an algebraic space, then F is an algebraic space. To do this, write $j_!F = U'/R'$ as in Lemma 0262 with $U', R' \in \mathrm{Ob}((Sch/S')_{fppf})$. Then the compositions $U' \to j_!F \to S$ and $R' \to j_!F \to S$ are morphisms of schemes over S'. Denote U, R the corresponding objects of $(Sch/S)_{fppf}$. The two morphisms $R' \to U'$ are morphisms over S and hence correspond to morphisms $R \to U$. Since these are simply the same morphisms (but viewed over S) we see that we get an étale equivalence relation over S. As $j_!$ defines an equivalence of categories of sheaves (see reference above) we see that $F = U/R$ and by Theorem 02WW we see that F is an algebraic space. □